EC 571: Digital VLSI Circuit Design

CMOS delay

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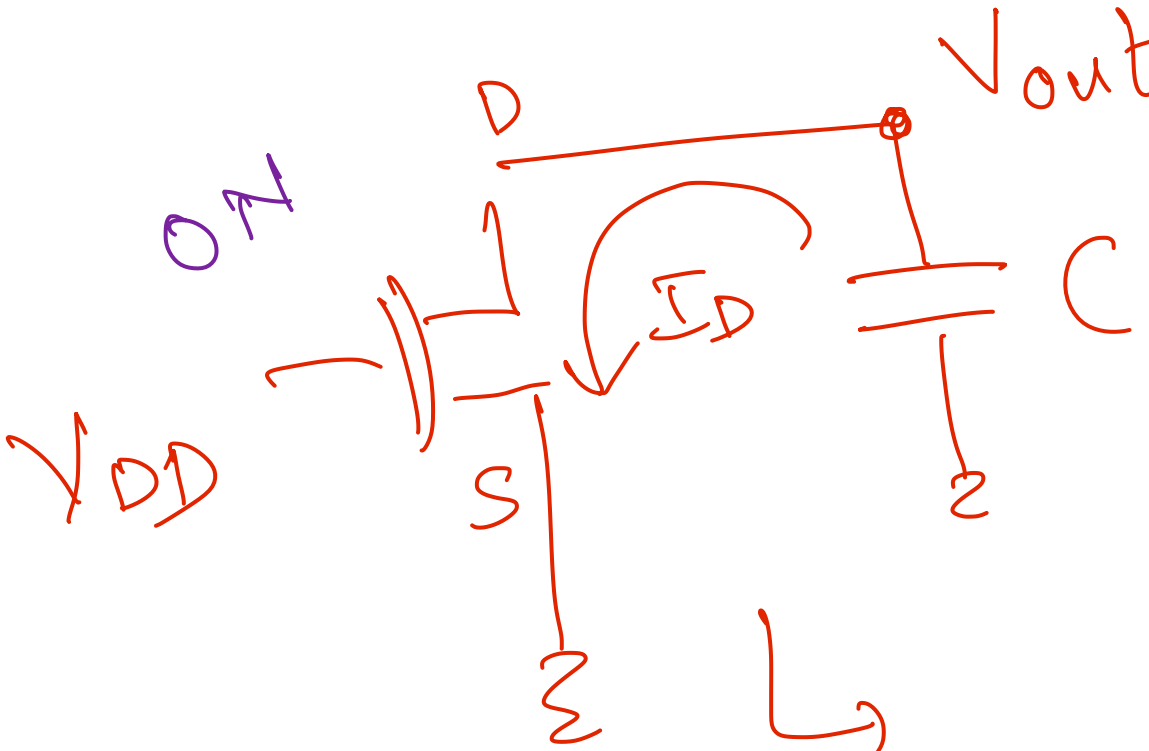
Acknowledgement - Some slides adapted from:
- J. Rabaey et al. Digital Integrated Circuits 2nd Ed.
- UC Berkeley EE141, UC Berkeley EE241, UC Berkeley EE251
- Stanford EE271

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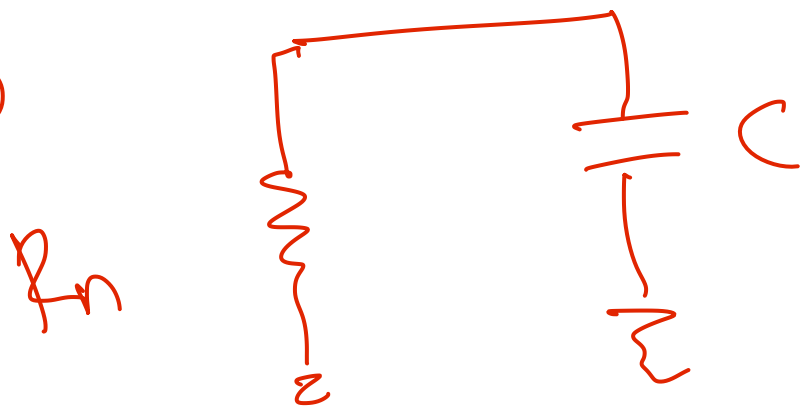
Last lecture

- Last lecture
  - Inverter VTC: Using MOS model
  - MOS Capacitance
- Today's lecture
  - CMOS delay
- Reading
  - Chapter 3,5

CMOS Switching delay.



$$I_D = C \frac{d V_{DS}}{dt}$$



$$t_p = \ln(2) R_n C.$$

Which I_D equation should we choose to get R_n?

Say → we are discharging the capacitance

$V_{DD} \rightarrow V_{DD}/2$

Find the time for this.

$V_{DS} = V_{DD} \rightarrow$ Saturation.

V_{DS} $V_{GS} - V_{TH}$

↓ ↓

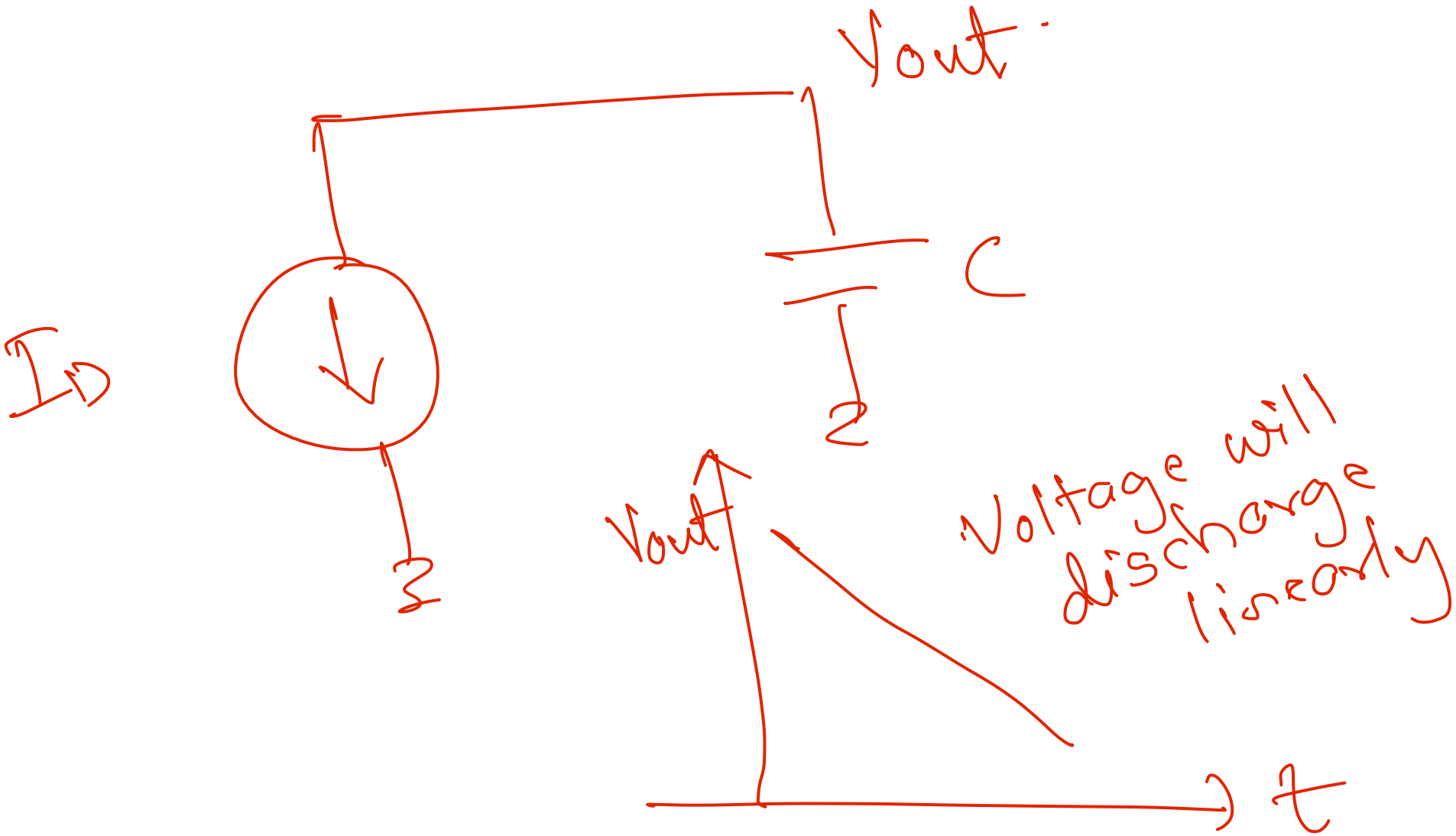
$V_{DD} > V_{DD} - V_{TH}$

As we start discharging we stay in saturation for most of time.

In saturation

$I_D \propto (V_{GS} - V_{TH})^2$

τ → depends on long-channel or short-channel device.



$$I = C \frac{\Delta V}{\Delta t}$$

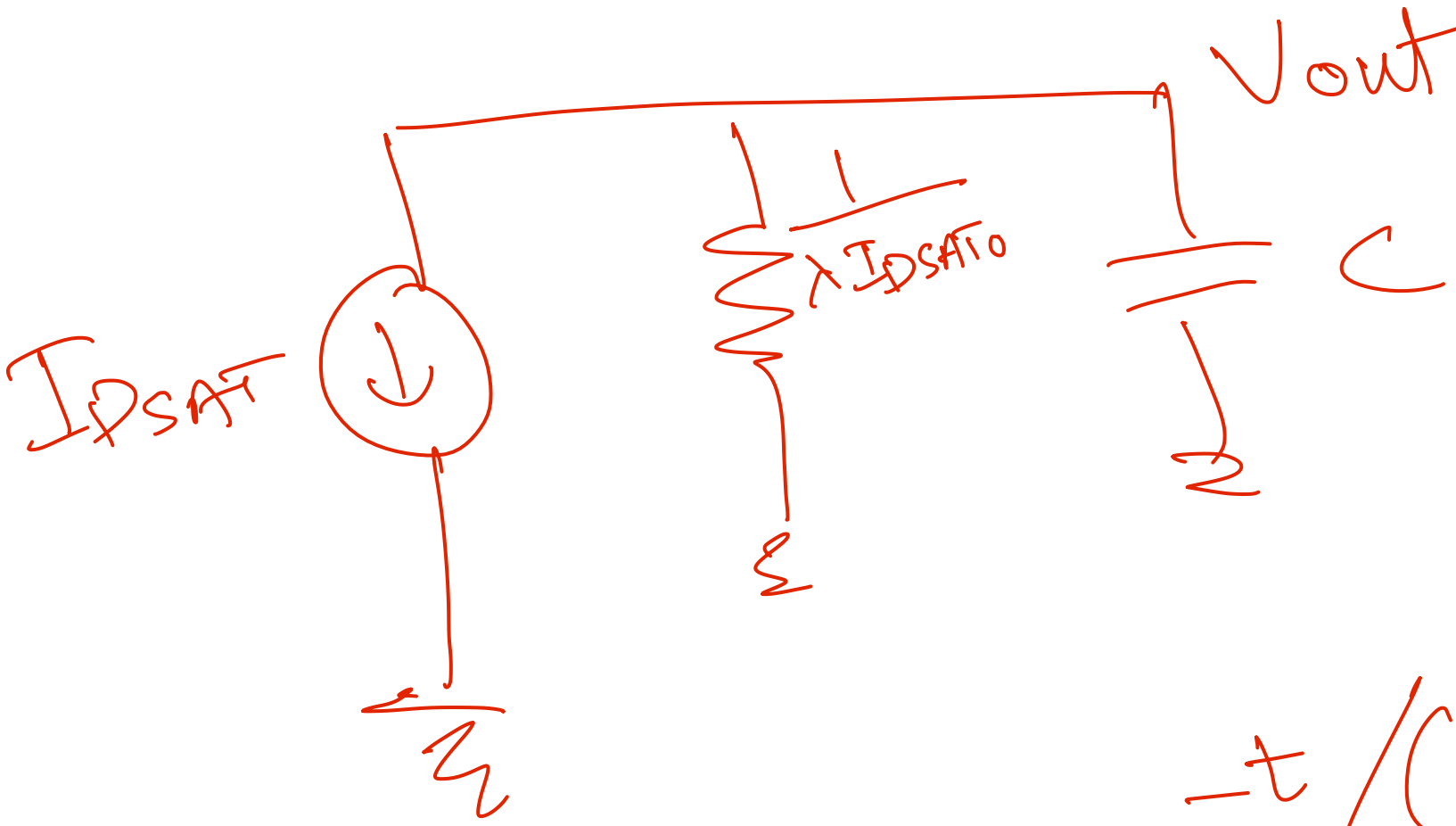
$$\Delta t = \frac{\Delta V \cdot C}{I}$$

$$= \frac{(V_{DD} - V_{DD}/2)\, C}{I_{D\,sat}}$$

Sidebar:

$I_{D\,sat}$ is not constant

$$I_{D\,sat} = I_{DSAT0}(1 + \lambda V_{DS})$$



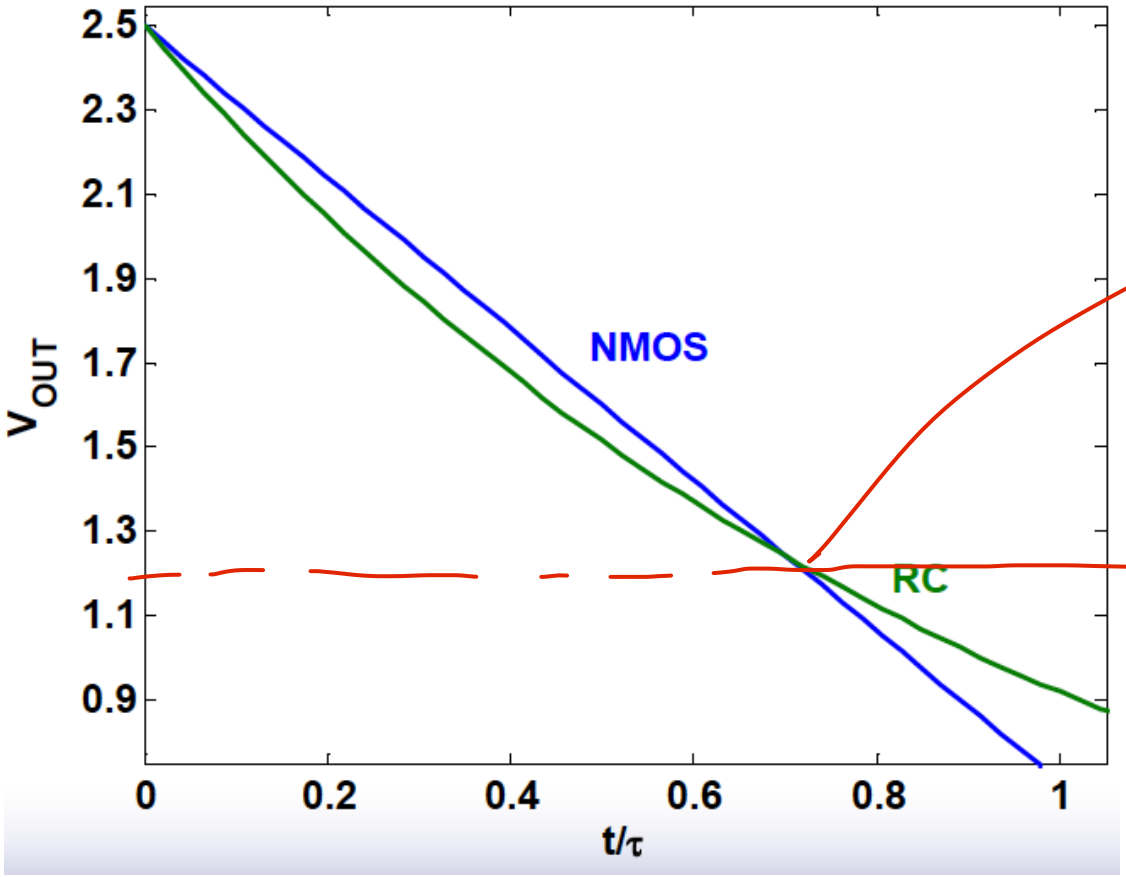
$$V_{out} = \left(V_{DD} + \frac{1}{\lambda}\right) e^{-t/(C/\lambda I_{DSAT0})} - \frac{1}{\lambda}$$

For small λ

$$t_p = \frac{C\left(V_{DD}/2\right)}{\left(1+\lambda V_{DD}\right) I_{DSAT0}}$$

RC Model

- Transistor current not linear on V_{OUT}
  - How is this going to work?



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$$\frac{C(V_{DD}/2)}{(1+\lambda V_{DD})I_{DSAT0}} = \ln(2) R_n C$$

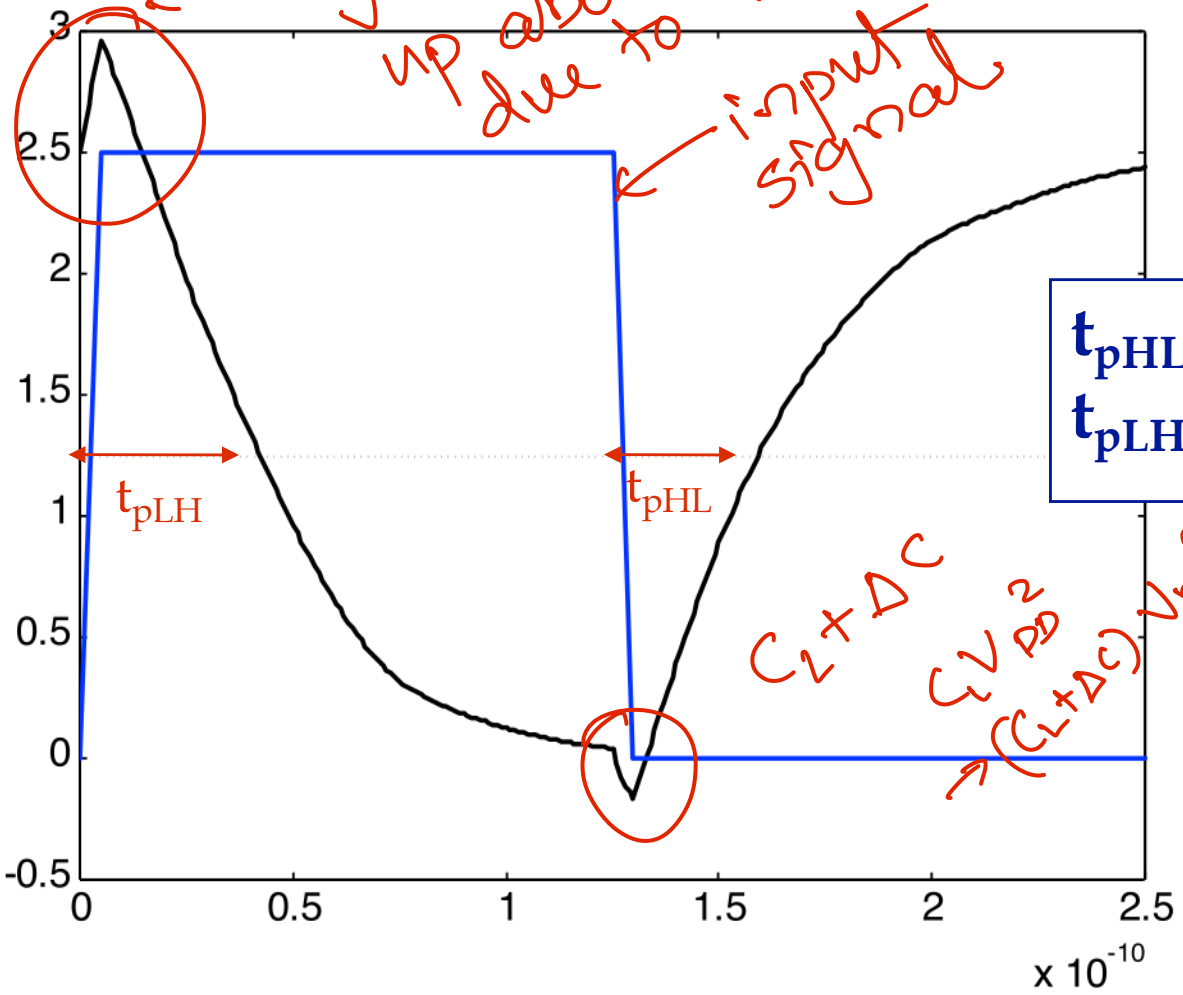
$$R_n = \frac{V_{DD}/2}{\ln(2)(1+\lambda V_{DD})I_{DSAT0}}$$

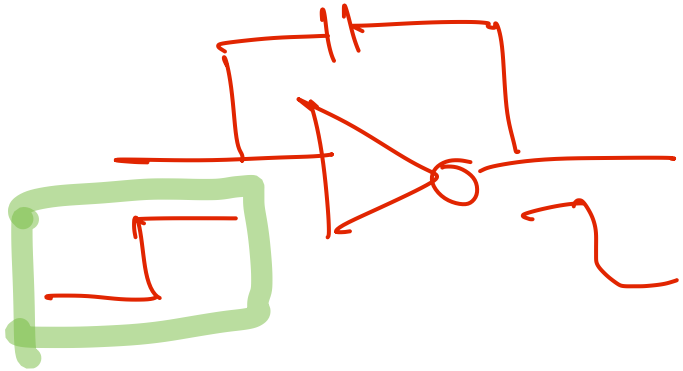
$$= \frac{V_{DD}}{2\ln(2) I_{DSAT0}}$$

(assuming λ is small)

$$R_n = 0.72 V_{DD} / I_{DSAT0}$$

Transient Response

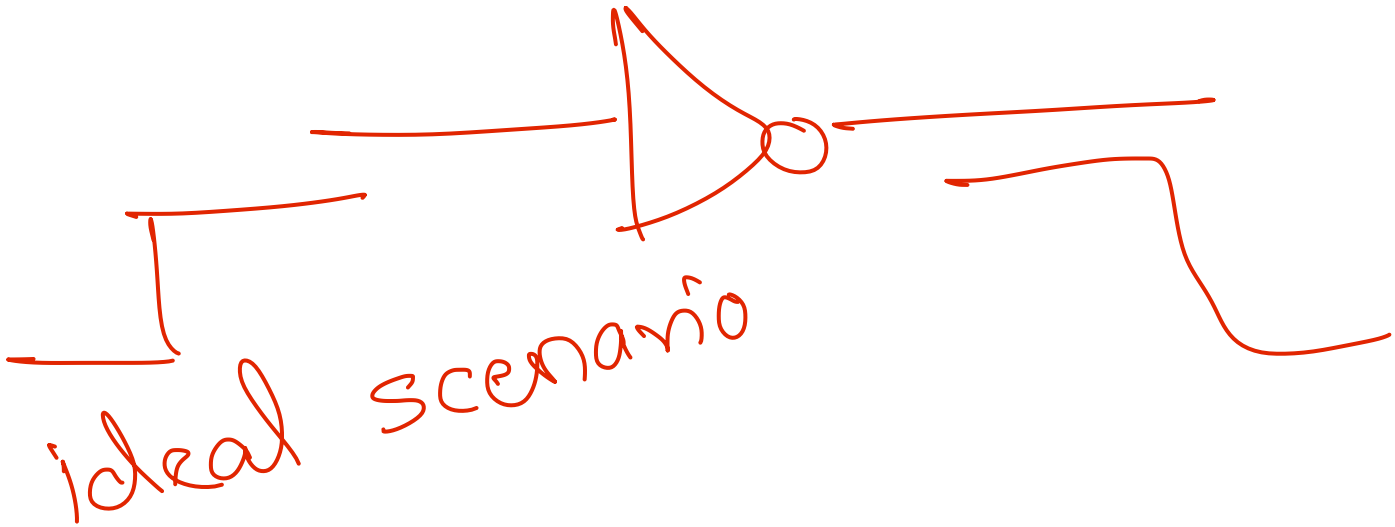


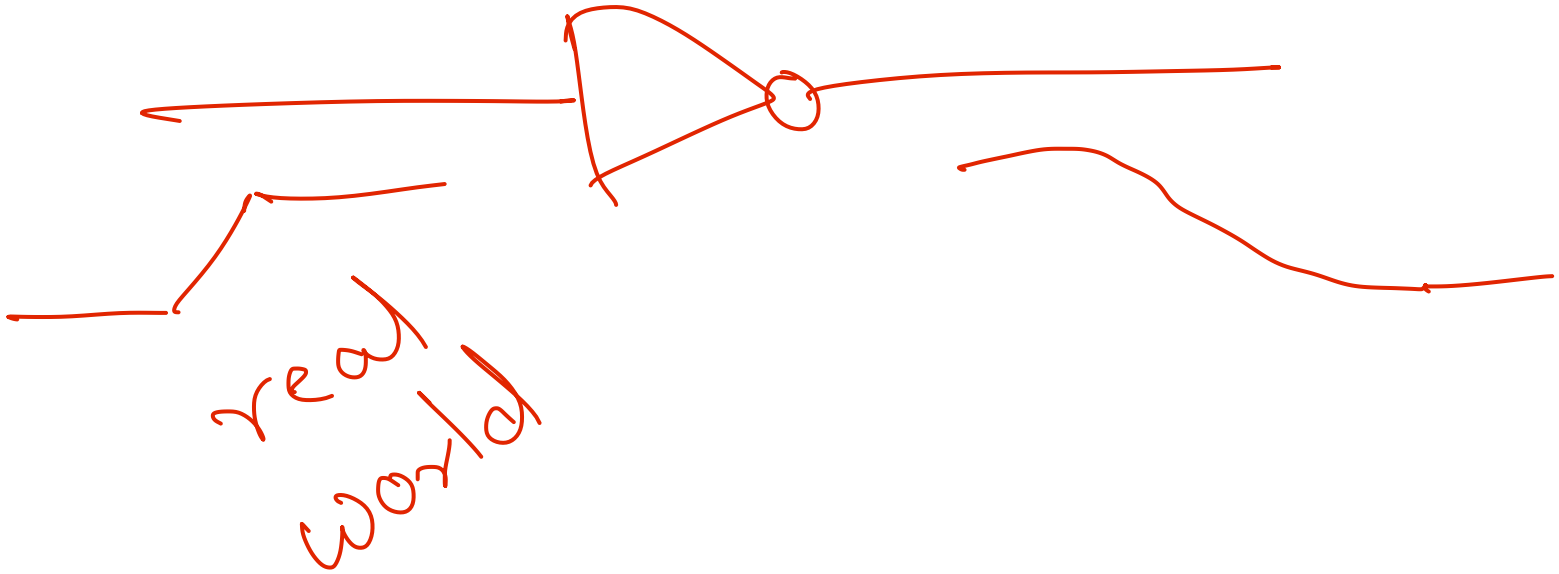


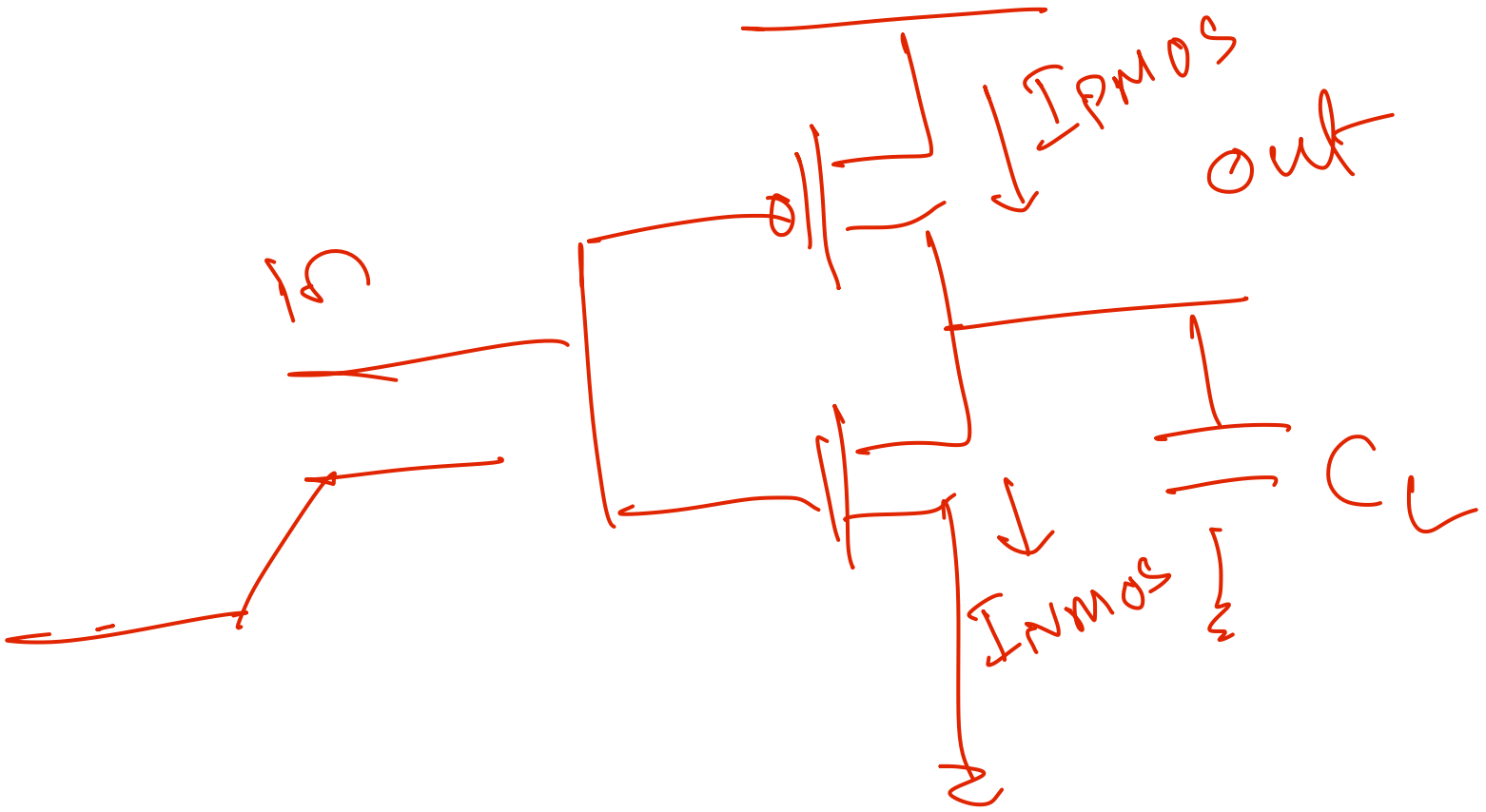
$$t_{pHL} = \ln(2)\, C_L\, R_{eqn}$$
$$t_{pLH} = \ln(2)\, C_L\, R_{eqp}$$

$$R_{eq} = V_{DD} / (2 \ln(2)(1 + \lambda V_{DD})\, I_{DSAT})$$

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ideal scenario


real world

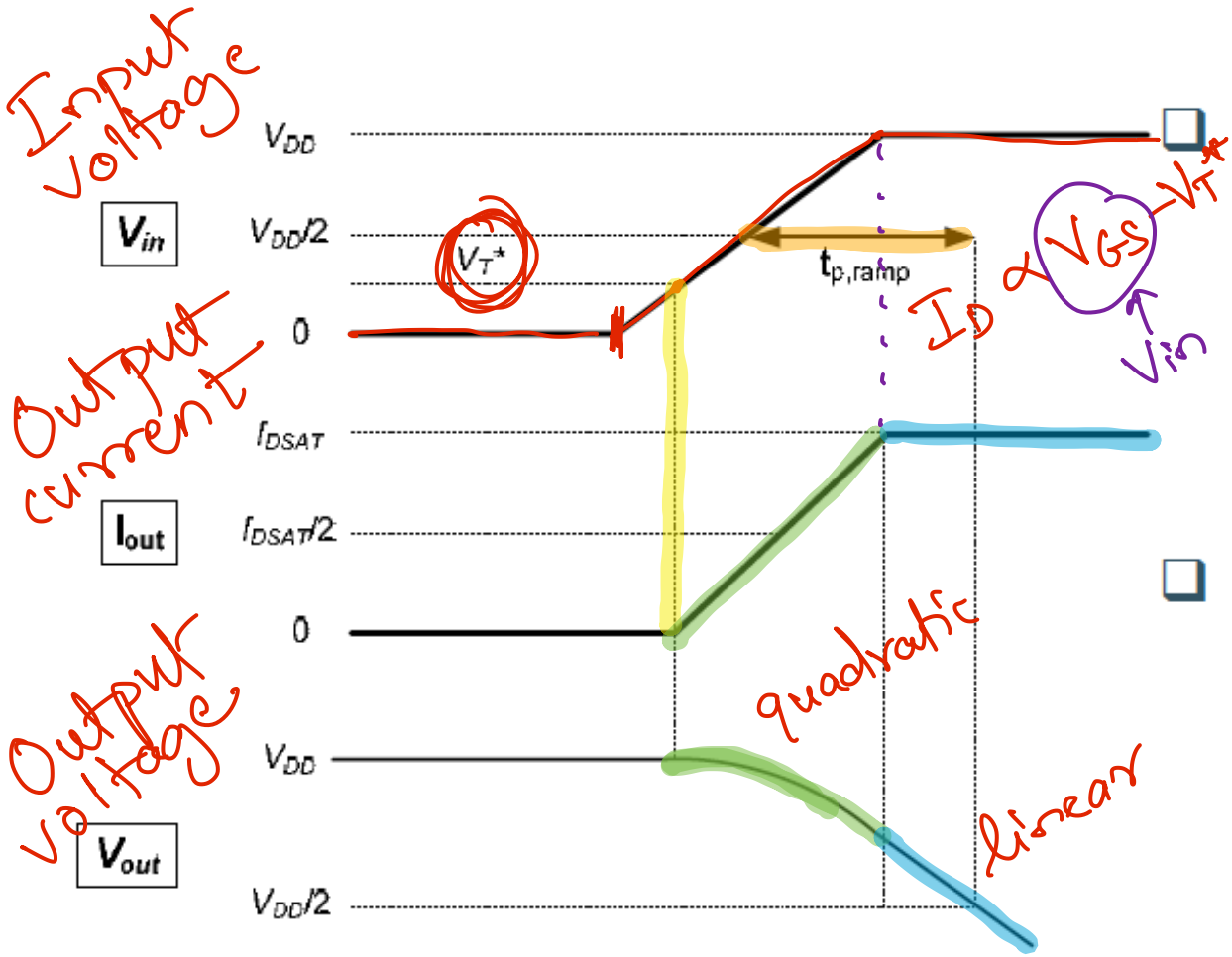


$$I_{PMOS} - I_{NMOS} = C_L \frac{dV_{out}}{dt}$$

I_{PMOS} → $V_{GS}(t)$

I_{NMOS} → $V_{GS}(t)$

Graphical analysis of slope dependence (1)

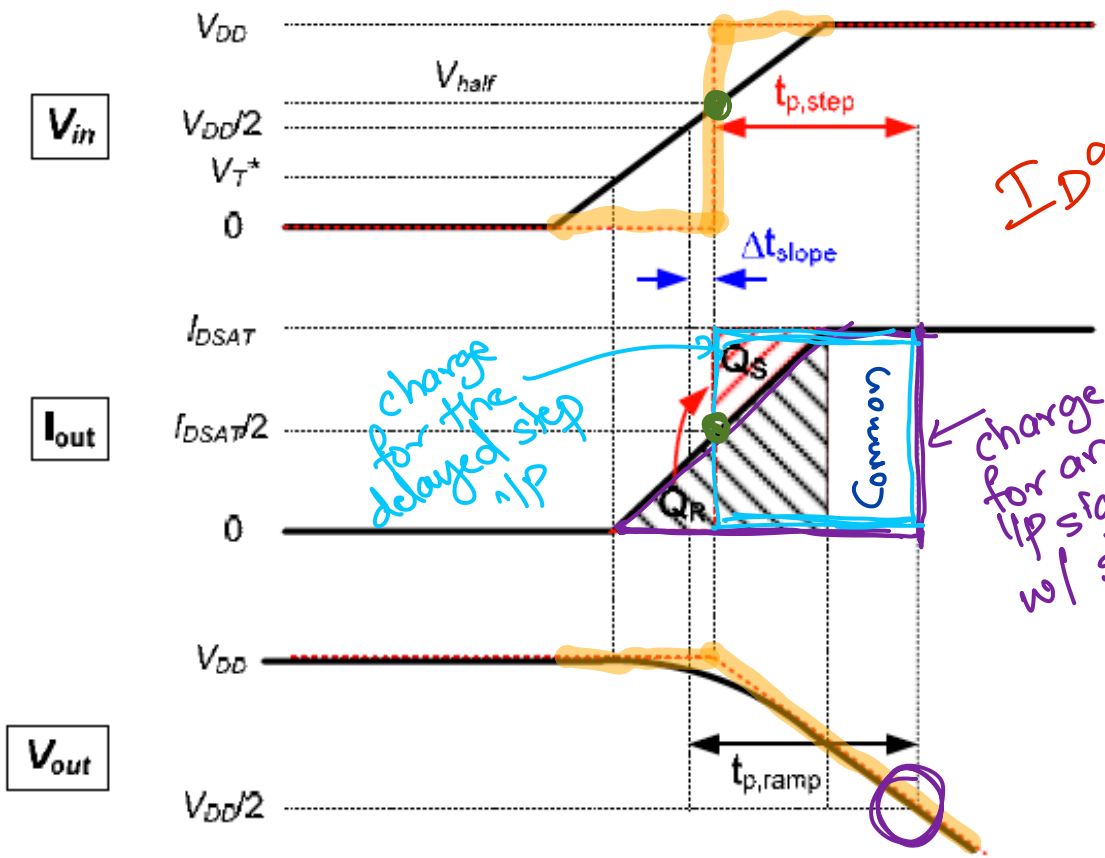


- I_{out} ramps linearly for $V_T^* < V_{in} < V_{DD}$
  - Constant once $V_{in} = V_{DD}$
- C_L integrates I_{out}
  - $V_T^* < V_{in} < V_{DD}$: V_{out} quadratic
  - $V_{in} = V_{DD}$: V_{out} linear

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Graphical analysis of slope dependence (2)

$t_p = \Delta t_{slope} + t_{pstep}$



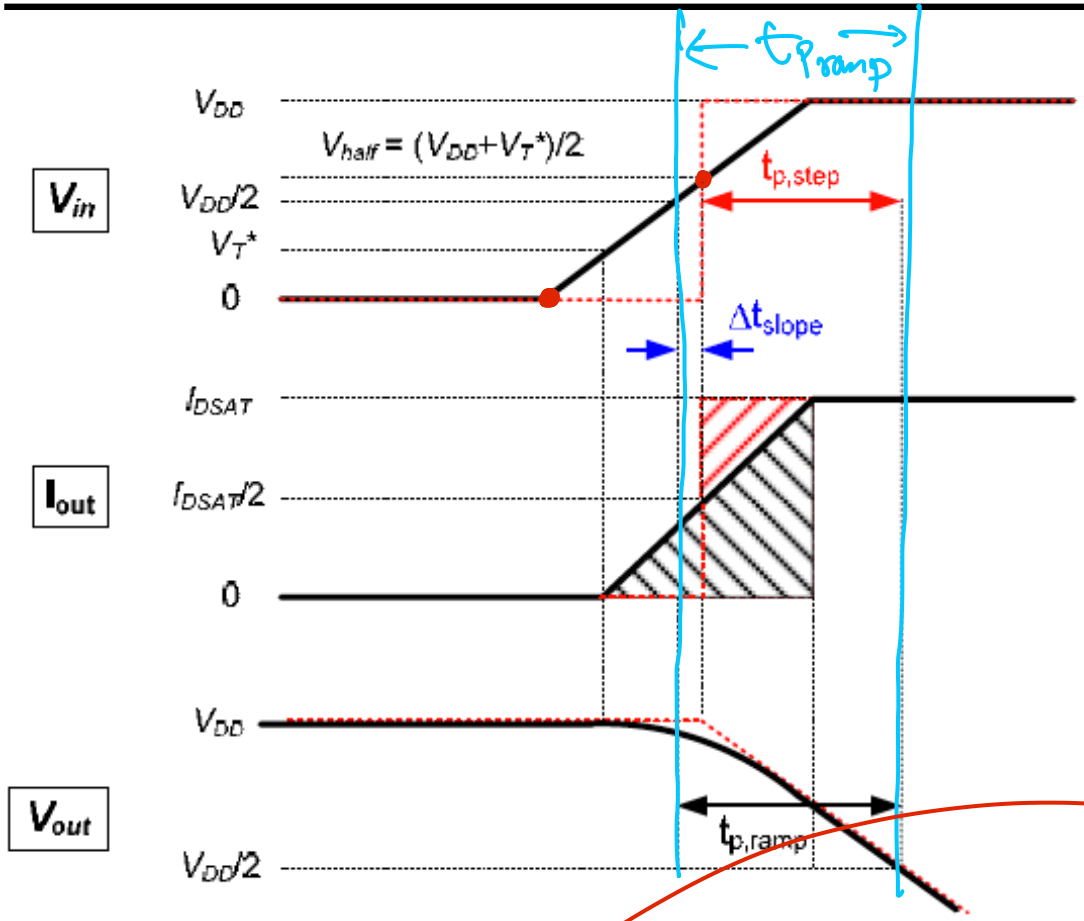
- Consider step input whose output crosses $V_{DD}/2$ at same time
  - delayed
  - $I_D \propto V_{GS} - V_{TH}$
- V_{out} set by charge removed from C_L
  - Need to make $Q_R = Q_S$
- Step has to shift to when $I_{out} = I_{DSAT}/2$



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$t_{pramp} = t_{pstep} + \Delta t_{slope}$

Graphical analysis of slope dependence (3)



$$t_{p,ramp} = t_{p,step} + \frac{V_T^*/2}{k_r} = t_{p,step} + \frac{V_T^*}{V_{DD}} t_{p,in}$$

- To find Δt_{slope}:
  - Find V_{in} when $I_{out} = I_{DSAT}/2$ (V_{half})
  - And use input t_r
- I_{DSAT} α (V_{DD}-V_T^*):

  $V_{half} - V_T^* = V_{DD}/2 - V_T^*/2$

  $V_{half} = (V_{DD}+V_T^*)/2$
- So $\Delta t_{slope} = (V_T^*/2)/k_r$
  - $k_r = V_{DD}/(2*t_{p,in})$

delay.

Using simple geometry.

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Find Δt_{slope}

$$I_{DSAT}(V_{half}) = \frac{I_{DSAT}(V_{DD})}{2}$$

$$I_{DSAT} \propto V_{GS} - V_T^*$$

$$V_{half} - V_T^* = \frac{V_{DD} - V_T^*}{2}$$

$$V_{half} = \frac{V_{DD} + V_T^*}{2}$$

Slope of i/p signal.

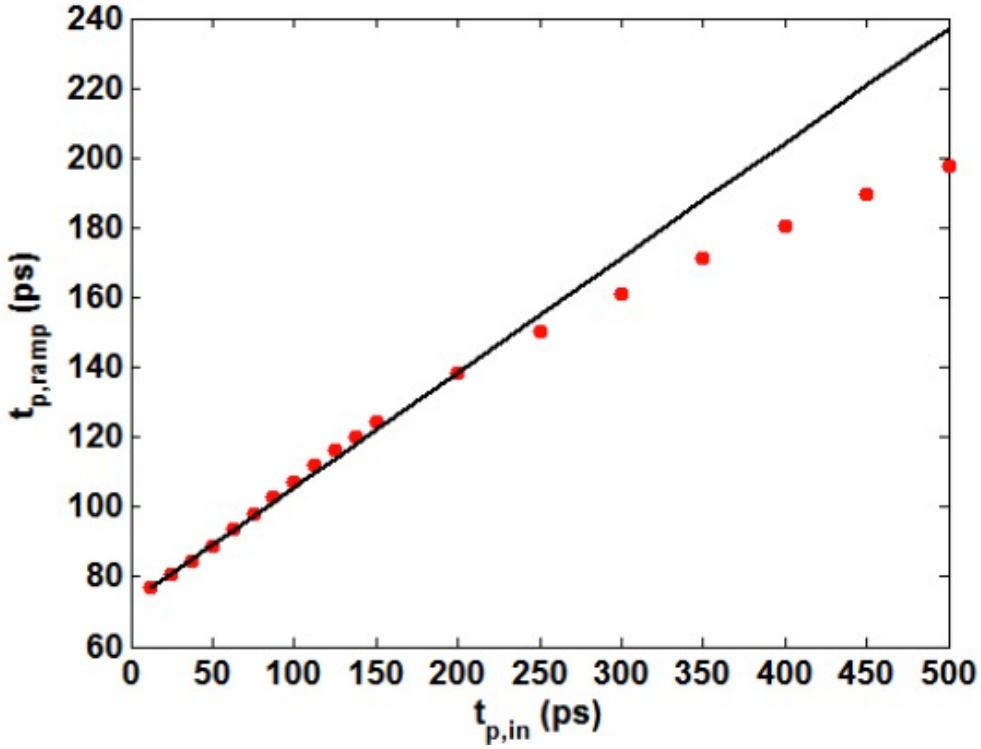
$$k_r = \frac{0 - V_{DD}/2}{0 - t_{pin}}$$

$$k_r = \frac{V_{DD}/2}{t_{pin}}$$

$$t_{Pramp} = t_{pstep} + \frac{V_T^*}{V_{DD}} t_{p\,i/p}$$

Model vs Spice data

- For reasonable input slope
  - Model matches well with Spice
- Model breaks with very large t_r
  - Input looks DC - traces VTC
  - Short-circuit current



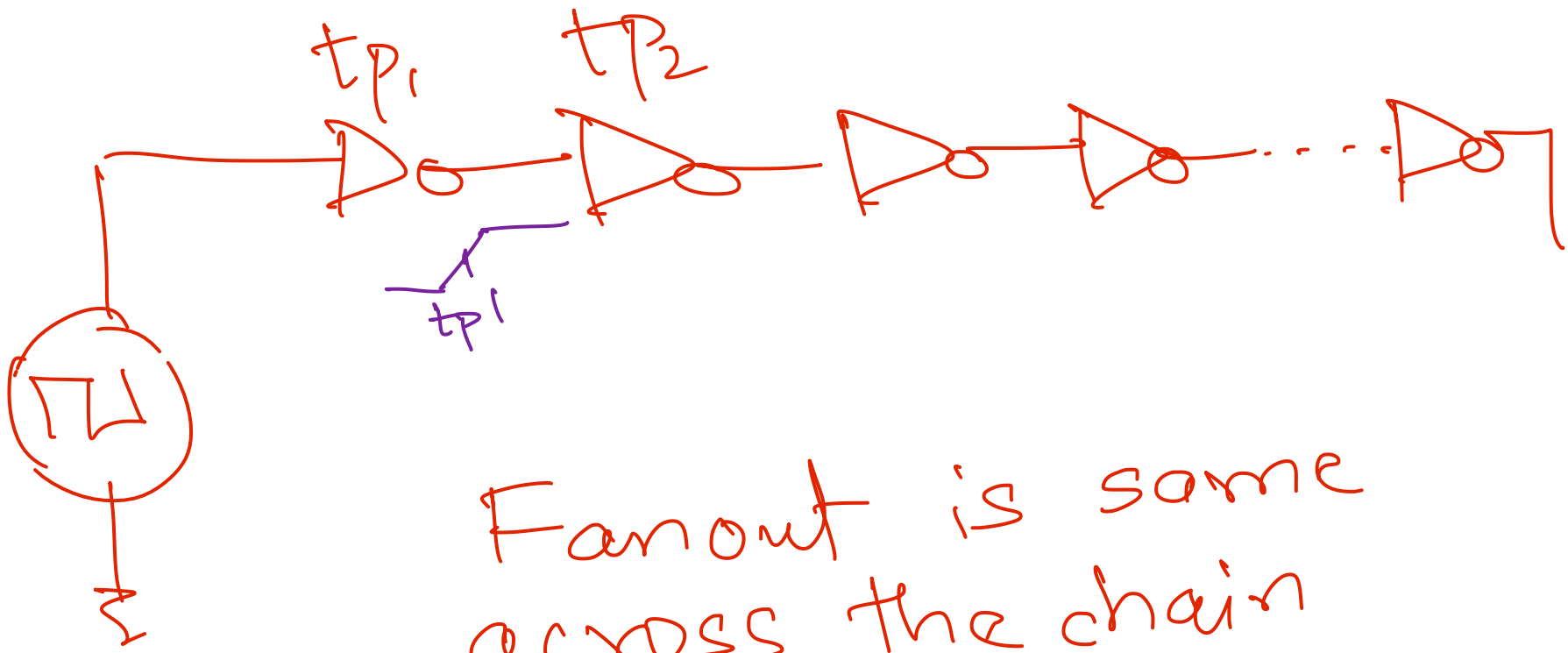
Result summary

- For reasonable input slopes

$$t_{p,ramp} = t_{p,step} + \frac{V_T^*}{V_{DD}} \cdot t_{p,in}$$

- For $t_{p,avg}$, V_T^* is $(V_{TN}^* + V_{TP}^*)/2$
  - V_{T^*}/V_{DD} typically ~ 1/3 at nominal supply

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Fanout is same across the chain

$$t_{p_1} = t_{pstep}$$

$$t_{p_2} = t_{pstep} + \frac{V_T^*}{V_{DD}} t_{p_1}$$

$$t_{p_2} = t_{pstep} + \frac{V_T^*}{V_{DD}} t_{pstep}$$

$$t_{p_2} = t_{pstep}\left(1 + \frac{V_T^*}{V_{DD}}\right)$$

$$t_{p_3} = t_{pstep} + \frac{V_T^*}{V_{DD}} t_{p_2}$$

$$t_{p_3} = t_{pstep}\left(1 + \frac{V_T^*}{V_{DD}} + \frac{V_T^{*2}}{V_{DD}^2}\right)$$

For large N

$$\frac{V_T^*}{V_{DD}} < 1$$

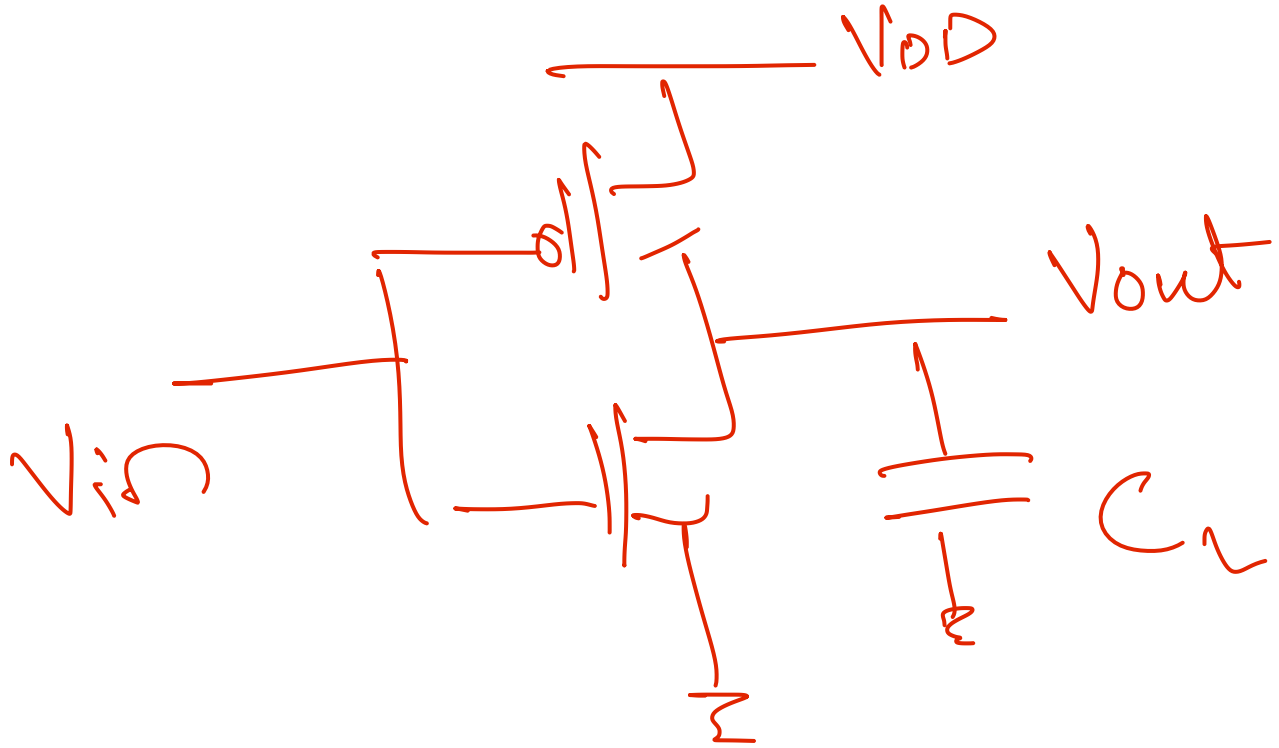
$$t_{pk} = t_{pstep} \left(\sum_{i=0}^{k-1} \left(\frac{V_T^*}{V_{DD}} \right)^i \right)$$

$$t_{pk} = \frac{\ln(2)\, RC}{1 - \frac{V_T^*}{V_{DD}}}$$

CMOS power dissipation

- Dynamic Power
- Static Power
- Leakage Power
- Short Circuit Power.

Dynamic Power

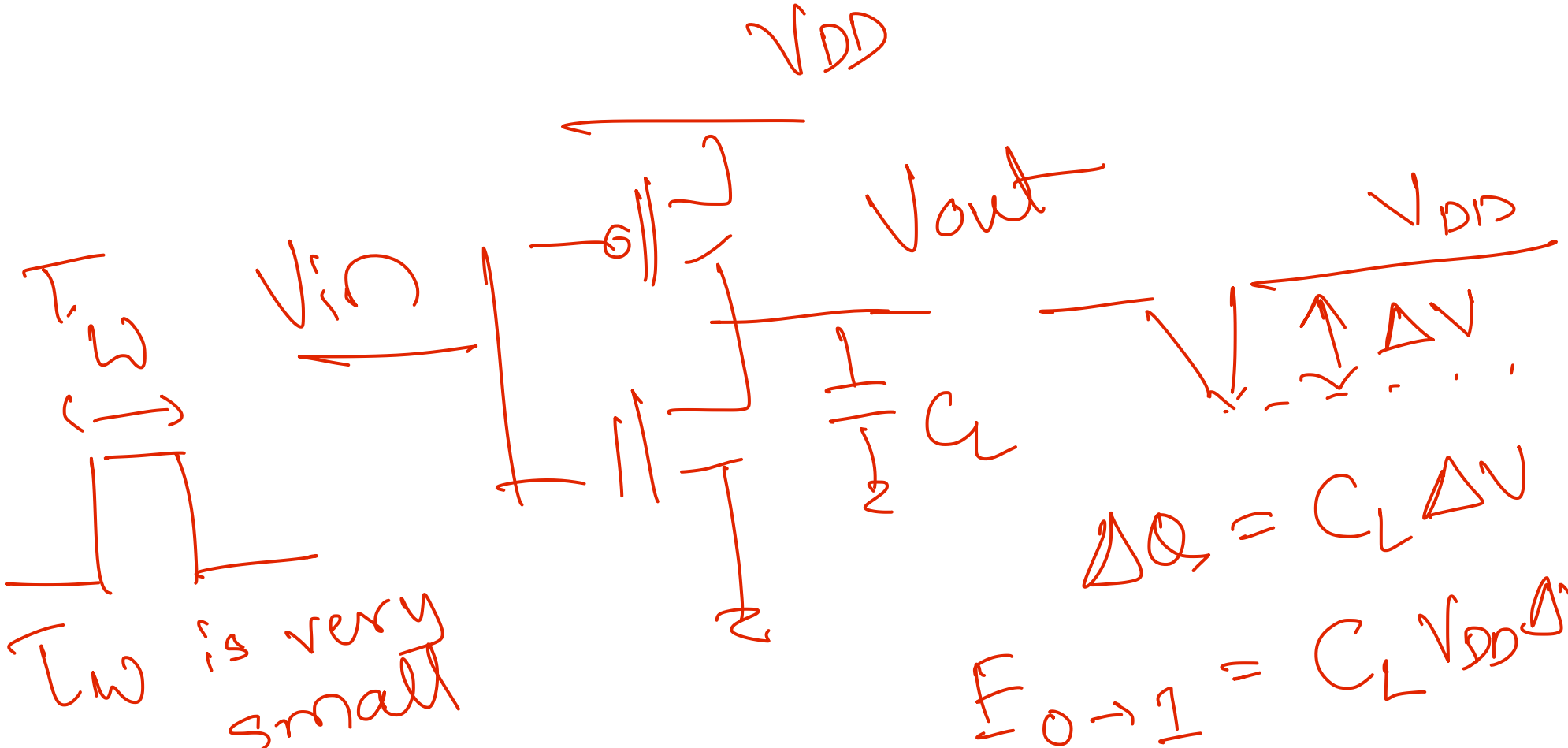


$$E_{0 \to 1} = C_L V_{DD}^2$$

$$E_C = \frac{1}{2} C_L V_{DD}^2$$

$$E_R = \frac{1}{2} C_L V_{DD}^2$$

$$E_{1 \to 0} = 0$$



$$\Delta Q = C_L \Delta V$$

$$E_{0 \to 1} = C_L V_{DD} \Delta V$$